

POSITIVE–UNLABELLED (PU) LEARNING

GRAPHS, ANOMALIES, AND THE ART OF LEARNING WITH INCOMPLETE INFORMATION

In the fast-evolving landscape of data science, the ability to learn from incomplete or uncertain data has become a challenge. Positive–unlabelled (PU) learning offers a promising solution: it trains algorithms to make reliable predictions even when only partial truth is known.

In many applications, crucial information is missing: a few fraudsters hide among millions of legitimate transactions, rare diseases go unrecorded in medical data and online communities form long before anyone notices them. Traditional algorithms that learn from clear examples of both good and bad cases are often unreliable and fail in these messy, incomplete realities.

ENTER PU LEARNING

That's where positive–unlabelled (PU) learning comes in. Instead of relying on perfect data, PU learning assumes we only know some of the positives, confirmed fraud cases for instance, while the rest of the data is unlabelled. Some of those unlabelled cases may also be fraudulent, but we can't directly observe them. Training reliable models under such uncertainty remains an ongoing challenge in modern data science.

A recent thesis named *Scalable Two-Step Imbalanced PU Learning with Graph Neural Networks* by Matthias Aerts and Romain Bérard addresses this challenge. The thesis

combines two exciting ideas in machine learning: graph neural networks (GNN) and anomaly detection.

Graphs allow computers to reason about relationships. They capture the notion of who trades with whom, which papers cite each other, which node is the central node in a network or how information flows through a social network. The iconic idea of PageRank is built on top of Graph Theory. In Machine Learning, graph applications have become very popular. The most common application of this field is a GNN. This is a neural network that can perform all the functions of a regular neural network, but on graphs. A celebrated application of a GNN is Deepmind's AlphaFold which predicts protein 3D structure from its amino acid sequence, the elementary protein building blocks.

MATHEMATICAL SPACE

In this thesis, GNNs are used not for biology but for financial and relational data. In short, a GNN transforms a complex web into a mathematical space where similar nodes, for instance users with comparable behavior, are positioned close together.

This mathematical or latent space is a lower-dimensional representation of the network. The latent space is a commonly used concept in machine learning to extract the most dominant features of data. First, the raw graph is processed to obtain the representations of nodes in the latent space. This low dimensional representation summarizes its local graph structure and features. To achieve this, a GraphSAGE-inspired architecture based on mini-batch training and sampling strategies is used. This means that the method can be used for large network sizes.

NEAREST-NEIGHBOUR ISOLATION FOREST

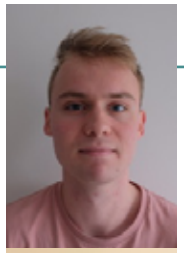
To filter out suspicious or ambiguous cases from these large graphs, the method uses a Nearest-Neighbour Isolation Forest. This technique is a combination of two data science techniques: Nearest-Neighbour and Isolation Forests. Isolation Forests use the premise that anomalous data points are easier to separate from the rest of the sample. The Nearest-Neighbour method exploits the possibility that fraudsters tend to occur in groups. In essence, fraudsters or anomalies are more separate from the usual behaviour and might come in groups. This yields a set of suggested positives and negatives. It leads to a cleaner dataset without needing to know how many bad actors exist.

The second step is then to use the node embeddings from GraphSage and the earlier obtained positives and reliable negatives to learn to predict which other nodes are positive among the unlabelled ones. This is done by XGBoost which draws the final line between whether nodes are likely positive or negative.

Graphs allow computers to reason about relationships.

IT PERSPECTIVE

From an IT perspective, this research opens up powerful new avenues. Its scalable design means it can process networks with hundreds of thousands of nodes, making it practical for deployment in operational environments. Think of financial institutions detecting illicit transactions in real time, or cybersecurity platforms identifying compromised devices before damage spreads.





Thesis title: Scalable Two-Step Imbalanced PU Learning with Graph Neural Networks

Authors: Matthias Aerts & Romain Bérard

Institution: KU Leuven

Why this thesis stands out
Aerts and Bérard's work bridges a gap between academic machine learning and enterprise-scale IT applications. It is an illustration of how intelligent systems can learn from incomplete data, adapt to complex relationships, and scale to the real world. It's a major step toward more robust, data-driven infrastructures in finance, healthcare, and beyond.

Key takeaways
- PU learning enables reliable predictions when only partial ground truth is available.
- Graph Neural Networks transform complex relational data into learnable structures.
- The two-step approach (anomaly filtering + classification) scales to enterprise-sized graphs.
- Applicable to fraud detection, cybersecurity, and healthcare analytics where data is incomplete.

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